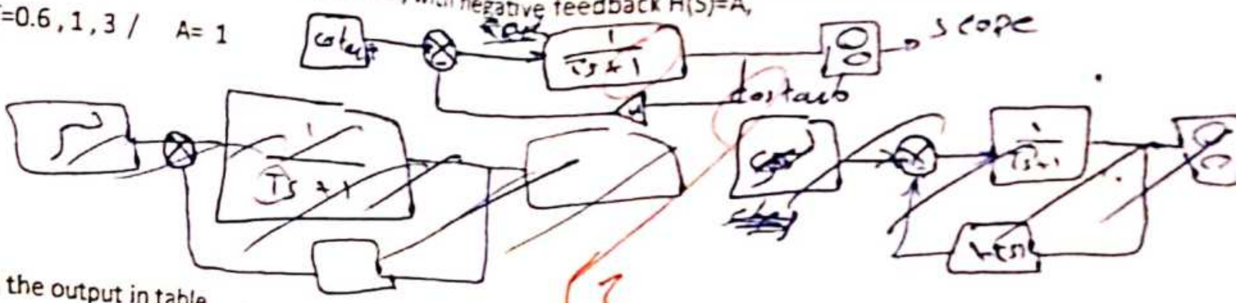
26/30

Albalqa'a Applied University
Mechatronics Engineering Dept.
Student ID:..............................

Faculty of Engineering technology
Control Theory Lab   Mid ExamA3
Name: [redacted]

Q1. a Simulate a first order of closed loop system, with negative feedback H(S)=A,

R(t) = 2 / T=0.6 , 1 , 3 / A= 1



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b. record the output in table

T	t(s)	0.5	1	1.5	2	2.5	3	3.5	4	4.5	5	5.5	6
0.6	C(s)	0.8	0.96	0.99	1	1	1	1	1	1	1	1	1
1	C(s)	0.63	0.86	0.95	0.98	0.99	1	1	1	1	1	1	1
3	C(s)	0.27	0.48	0.63	0.74	0.81	0.86	0.9	0.93	0.95	0.96	0.97	0.98

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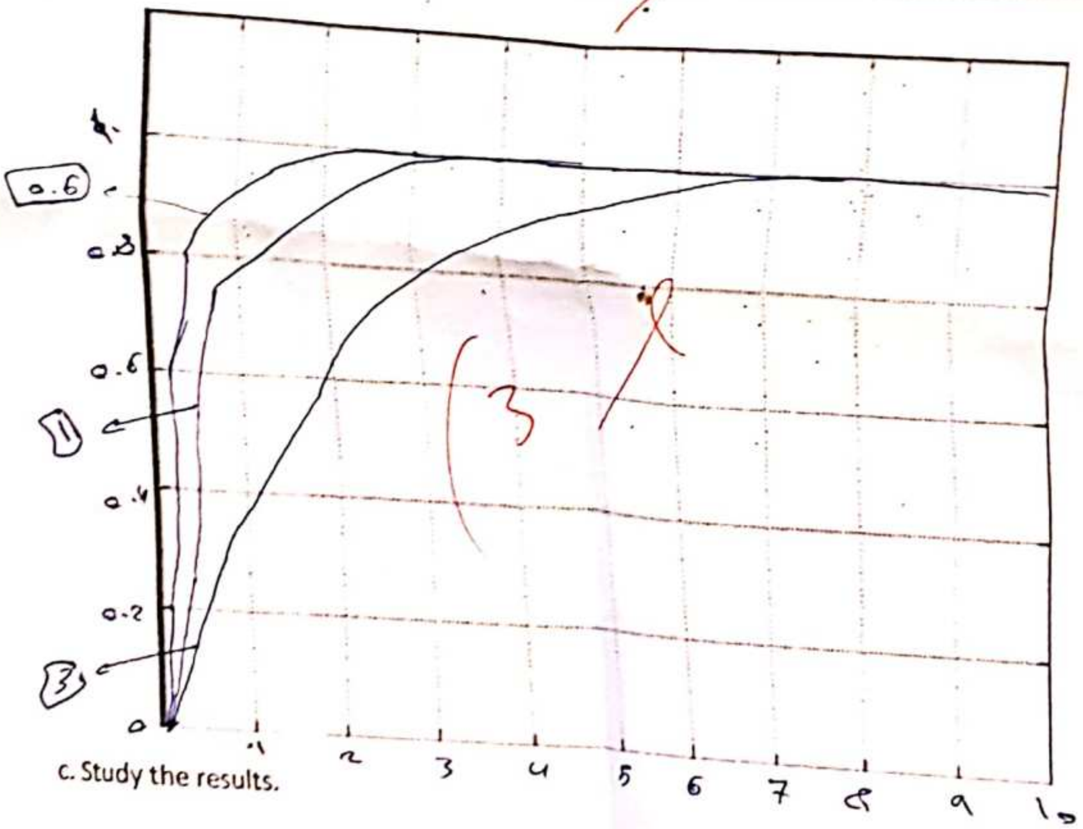
c. Plot the output



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c. Study the results.

كلما زاد T كلما قل الوصول إلى حاله steady state بشكل أسرع

مع ثبات feedback

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Al-Balqa'a Applied University
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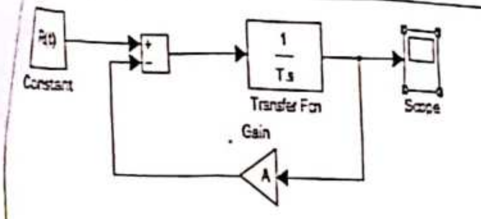
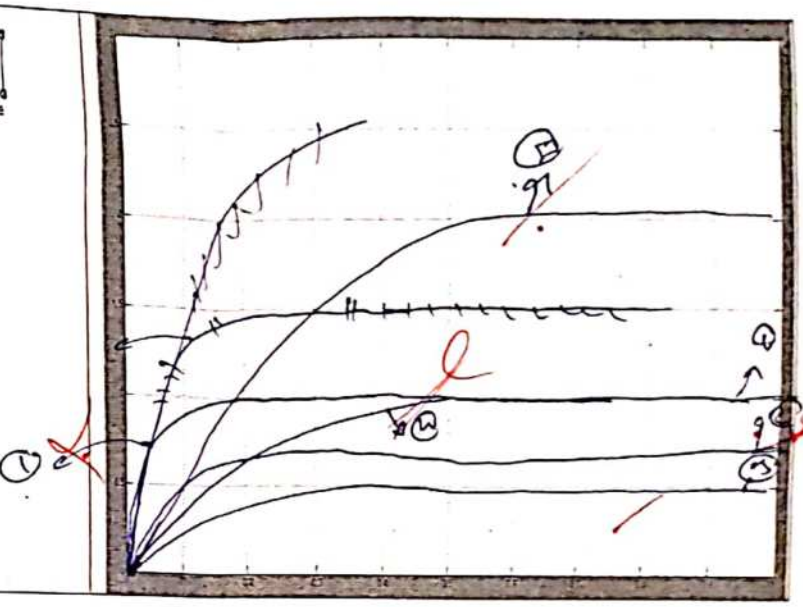
Faculty of Engineering technology
Control Theory Lab Mid ExamA5
Name:.....

Q2. Plot the output response for each of the following conditions



1. R(t)=1, T=1, A=0
2. R(t)=1, T=1, A=1
3. R(t)=1, T=1, A=3
4. R(t)=1, T=2, A=1
5. R(t)=2, T=1, A=1

(6



Q3. For a system given where Y(t) is system output and m(t) is system input ; r(t) = 2

$$y'''(t) - 3y''(t) + 2y'(t) + 4y(t) = r(t)$$

$$y'''(t) = 3y''(t) - 2y'(t) - 4y(t) + r(t)$$

$$x_1 = y \rightarrow x_1' = y' = x_2$$

$$x_2 = y' \quad x_2' = y'' = x_3$$

$$x_3 = y'' \quad x_3' = y''' = 3x_3 - 2x_2 - 4x_1 + 2$$

$$\begin{bmatrix} x_1' \\ x_2' \\ x_3' \end{bmatrix} = \overset{A}{\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -4 & -2 & 3 \end{bmatrix}} \begin{bmatrix} x_1 \\ x_2 \\ x_1 \end{bmatrix} + \overset{B}{\begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix}} [r(t)]$$

$$[y] = \overset{C}{[1 \quad 0 \quad 0]} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \overset{D}{[0]} [r(t)]$$

a. Determine the state space matrices

Parameters

A:
[0 1 0; 0 0 1; -4 -2 3]

B:
[0; 0; 2]

C:
[1 0 0]

D:
[0]

7.5

b. Solve this equation by the transfer function

$$s^3 y(s) - 3s^2 y(s) + 2s\, y(s) + 4 y(s) = r(s)$$

$$y(s)(s^3 - 3s^2 + 2s + 4) = r(s)$$

$$\boxed{\frac{y(s)}{r(s)} = \frac{1}{s^3 - 3s^2 + 2s + 4}}$$